

Mohammed Afaan Ansari

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AI/ML Engineer | Generative AI, Multimodal LLMs & RAG. Cut LLM cost 99.98% & inference 40% at MSCI; 5 peer-reviewed papers; ASONAM 2023 *2nd Best Paper*; lead author COLM 2026 (84%).

EDUCATION

University of Maryland, College Park

M.S. Computer Science | GPA: 3.81/4.0 | Advisor: Prof. C. Buntain (CLIP/iSchool)

Aug 2024 – May 2026

College Park, MD

Sardar Patel Institute of Technology, Mumbai University

B.Tech. Computer Engineering | GPA: 8.9/10 | ACM/Springer publications: NLP & blockchain

Jul 2019 – Jun 2022

Mumbai, India

TECHNICAL SKILLS

- **LLMs / GenAI:** PyTorch, HuggingFace, LoRA/PEFT, vLLM, ONNX, TorchScript; fine-tuned Phi-3, Qwen-3, InternVL2.5, LLaVA, Mistral, Gemma; RAG (LangChain, LlamaIndex); BERT, RoBERTa, GPT-4, CLIP; quantization, mixed-precision, GPU profiling, RLHF basics.
- **MLOps & Cloud:** Python, SQL, PySpark; GCP (Vertex AI, BigQuery), AWS (SageMaker, S3); Docker, Kubernetes, Airflow, GitHub Actions CI/CD, Terraform; REST APIs, scikit-learn, pandas, NumPy.

EXPERIENCE & RESEARCH

CSSI Inc.

Data Science Intern

Jun 2025 – Aug 2025

Washington, D.C.

- Architected anomaly-detection pipeline on 10M+ FAA aviation-weather records (90%+ recall), enabling analysts to surface hazardous flight patterns 2× faster.

MSCI Sustainability Institute

Research Participant — Green CapEx Initiative

Jul 2025 – Sep 2025

Remote

- Computed ESG/CapEx metrics from 10K+ records adopted into MSCIT's Green CapEx taxonomy used by institutional investors.

University of Maryland — CLIP Lab & iSchool

Graduate Research Assistant — Multimodal LLMs (Prof. C. Buntain)

Jan 2025 – Present

College Park, MD

- Benchmarked 9 vision-language model families (InternVL2.5, Gemma 3 27B, LLaVA 1.6, Pixtral 12B, Phi-3.5 V) on crisis fact extraction; chosen model produced 50% high-importance facts at 10× lower runtime (28s vs 295s/image).
- Shipped multimodal RAG (LVLM extraction → Mistral-7B summarization → embedding-dedup) and trained RoBERTa-large priority classifier on 126K labeled posts (macro-F1 0.77); deployed for 20+ analysts, cutting turnaround 2h → 15m.

MSCI Inc.

Data Science Technology Associate | Star Performer 2023, top of 1,200+

Jul 2022 – Aug 2024

Mumbai, India

- Promoted to Associate in 1yr (vs. typical 3–4); led GenAI extraction on 100K+ financial reports (90%+ accuracy vs. 60% manual); cut serving cost \$10→\$0.002/1K (**99.98% reduction**) via prompt-engineering + ONNX (<100ms); powered MSCI's first hybrid-cloud deployment.

Georgia Tech & Mayo Clinic

Research Assistant — ML & LLM Systems (Prof. Joy Arulraj)

Dec 2022 – Dec 2023

Atlanta, GA (Remote)

- Built CNN-based MRI tumor-detection (85% accuracy on 1K patients) targeted for clinical integration at Mayo Clinic; benchmarked GPT-4 vs. ONNX-quantized vision encoders within EVADB, cutting per-image latency 40% (150ms → 90ms).

Rochester Institute of Technology

Research Assistant — Social Data Mining (Prof. A. KhudaBukhsh)

Jul 2022 – Jun 2023

Rochester, NY (Remote)

- Co-authored ACM/IEEE paper measuring social-web dataset transience using BERT on Twitter/Reddit corpora; *2nd Best Paper, ASONAM 2023*.

Sourceable Inc. (Columbia-backed startup)

Lead Software Developer

Jan 2022 – Jul 2024

New York, NY (Remote)

- Led full-stack development (React, Node.js, Docker, K8s) of crisis-mapping platform adopted by 1,500+ journalists across 52+ countries; cut page load 30% and lifted retention 25%.

Nomura Holdings — Global Markets IT

Technology Analyst Intern

Jan 2022 – Jun 2022

Mumbai, India

- Engineered ELK-stack monitoring for trading infrastructure (incident resolution -40%); secured full-time return offer with 40% salary uplift vs. incoming batch.

SELECTED PROJECTS & LEADERSHIP

- **ChartDesign (COLM 2026, lead author)** — LoRA-tuned LLMs (CSV → chart JSON); 84% accuracy vs. 55% zero-shot; 65% preferred over human charts.
- **ARIA Risk Intelligence Agent** (sole author) — 10-agent Claude + FastAPI + React; 70–80% CVE triage reduction; mentored 40+ students.